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8.2.2 Toward practical descent methods

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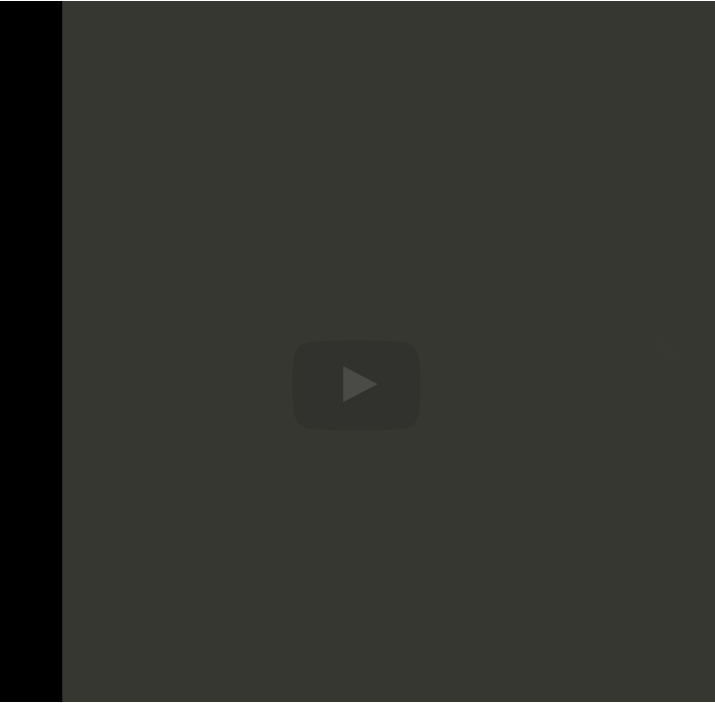
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.....

vector multiplies and a bunch of dot products." Now depending on how many non-zeroes there are in our matrix A, it's usually the case that it's the matrix vector multiply where quite a bit of the expense is. So reducing the number of matrix vector multiplies that we have to do is a good thing. And the way we

have it formulated right now is we need one matrix vector multiply right here.

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Reading assignment

1/1 point (graded)



Read Unit 8.2.2 of the notes. [\[LINK\]](#)

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Homework 8.2.2.1

1/1 point (graded)
Let

$$\begin{aligned}x^{(k+1)} &= x^{(k)} + \alpha_k p^{(k)} \\ r^{(k)} &= b - Ax^{(k)} \\ r^{(k+1)} &= b - Ax^{(k+1)}\end{aligned}$$

Show that

$$r^{(k+1)} = r^{(k)} - \alpha_k Ap^{(k)}.$$

☒ done



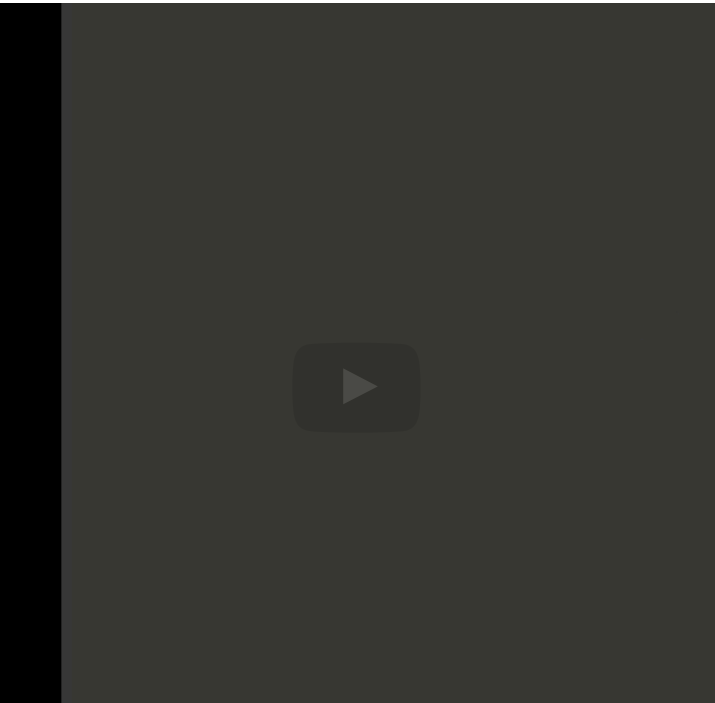
Solution:

For a complete solution see the [notes](#).

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this requires you to compute A times p^(k) and this requires you to compute A times p^(k). So once we've actually determined what the direction is, what we can do is right after that compute vector q^(k) equal to A times p^(k). And once we do that we can replace this right here with q^(k), and replace this right here with q^(k). And what we notice is that now we have an algorithm that only requires one matrix

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Homework 8.2.2.2

1/1 point (graded)
For loops in the algorithm in

```
Given :  $A, b, x^{(0)}$ 
 $r^{(0)} := b - Ax^{(0)}$ 
 $k := 0$ 
while  $r^{(k)} \neq 0$ 
   $p^{(k)} :=$  next direction
   $q^{(k)} := Ap^{(k)}$ 
   $\alpha_k := \frac{p^{(k)T} r^{(k)}}{p^{(k)T} q^{(k)}}$ 
   $x^{(k+1)} := x^{(k)} + \alpha_k p^{(k)}$ 
   $r^{(k+1)} := r^{(k)} - \alpha_k q^{(k)}$ 
   $k := k + 1$ 
endwhile
```

count the number of matrix-vector multiplications, dot products, and "axpy" operations (not counting the cost of determining the next descent direction).

1

✓

matrix-vector multiplies (mvmults)

Answer: 1

2

✓

dot products (dots)

Answer: 2

2

✓

"axpy"s

Answer: 2

Solution:

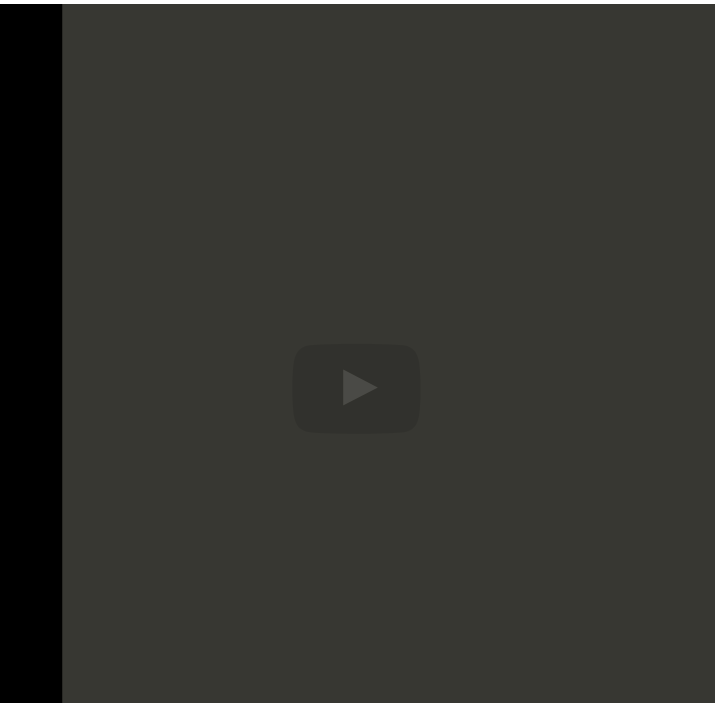
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And that is that actually we can formulate this algorithm without storing all of the intermediates vectors. We can just overwrite vectors x , r , q , and the value α , as long as when we compute our next direction, we don't actually need to know about the history of all of the vectors that were computed before it. And that turns out to be key to keeping the memory requirements of these algorithms down

2.0x

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